Korea:

The state of High Temperature Superconductivity in 2025

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Cooling System at Seonyu Substation for the Munsan-Seonyu 23kV HTS Platform. *Credit © KEPCO Research Institute*

Energy Context Highlights

Korea's power system is facing new challenges due to the growth of large-scale electricity demand from data centers, semiconductor facilities, industrial complexes, and renewable energy projects. In urban and load-dense areas, it is becoming more difficult to secure sites for new substations, install additional underground cables, shorten construction periods, and manage public acceptance issues.

High-temperature superconducting (HTS) power technology can be one possible solution to these challenges. HTS cables can transmit large amounts of power in a compact space. Therefore, they can be useful in areas where land, ducts, or existing power facilities are limited. Korea is promoting the Munsan-Seonyu 23 kV HTS Platform Demonstration Project as a key step toward practical grid application and future business development.

3-Phase Concentric HTS Cable System at Seonyu Substation. *Credit © KEPCO Research Institute*

Policy/Regulatory/Infrastructure Highlights

To apply HTS power technology to the actual power grid, technical development alone is not enough. Design standards, testing procedures, operation guidelines, and maintenance frameworks are also needed. An HTS cable system is not a cable-only facility. It is an integrated power system that includes a cooling system, vacuum insulation, monitoring equipment, and protection and control systems.

The Munsan-Seonyu superconducting platform demonstration project is a key project for building this foundation. The project is scheduled for completion in October 2026. The buildings have already been completed. Cable installation and cooling system installation have also been completed. The HTS cable system is now about 85% complete. At present, vacuum and evacuation work is being carried out to prepare for initial pressurization and commissioning.

The next steps include cooling operation, electrical testing, protection coordination review, grid connection tests, and commissioning. The experience gained from this project will support future design standards, testing procedures, operation guidelines, and maintenance rules for HTS power systems.

R&D Activities Highlights

Korea's R&D activities are moving from component-level development to system-level grid application. The Munsan-Seonyu project is being used to verify not only the HTS cable, but also the cooling system, protection and control system, monitoring system, and grid operation method.

The main technical items include cable performance, insulation performance, cooling stability, vacuum condition, thermal load management, fault response, and recovery procedures. These items are important for the long-term reliable operation of HTS power equipment in the actual grid.

Future R&D topics include 154 kV-class superconducting fault current limiters (SFCLs), MVDC HTS cables, and asset management-based operation technologies for HTS power systems. A 154 kV-class SFCL can help reduce fault current and lower the burden on existing substations and circuit breakers. MVDC HTS cables can be considered for renewable energy integration, industrial complexes, and urban load-dense areas.

KEPCO is also reviewing the connection between HTS power systems and asset management technologies to support wider use in private-sector applications such as data centers. Unlike mature commercial power equipment, it may be difficult to keep all major spare parts for HTS cable systems available at all times. Therefore, it is important to monitor the condition of key facilities such as circulation pumps, chillers, cooling system components, and conventional power equipment connected to the HTS system. By combining HTS operation data with KEPCO's asset management technologies, it will be possible to predict maintenance and replacement timing, prevent failures, support performance assurance, and develop operation service models for private customers.

Demonstration Projects Highlights

The Munsan-Seonyu superconducting platform demonstration project is one of Korea's key HTS power demonstration projects. The purpose of the project is to verify a 23 kV HTS cable-based power supply system in the actual power grid.

This project has broader meaning than the previous Singal-Heungdeok HTS cable demonstration project. The Singal-Heungdeok project showed the possibility of operating an HTS cable in the actual power grid. The Munsan-Seonyu project goes one step further by testing HTS power technology as a new power supply platform.

In particular, the Munsan-Seonyu superconducting platform is meaningful because it can suggest an alternative model for urban areas where it is difficult to build a new 154 kV substation. In cities and load-dense areas, it is often difficult to secure substation sites, install additional underground cables, shorten construction periods, and manage public acceptance issues. In such cases, a superconducting platform can be considered as a compact and high-capacity power supply option using a 23 kV-class system.

The project is scheduled for completion in October 2026. All buildings have been completed. The HTS cable and cooling system have been installed. The HTS cable system is about 85% complete. The current work is focused on vacuum and evacuation processes for initial pressurization and commissioning.

The operation data and field experience from this project will be used for future project development. Candidate applications include data centers, industrial complexes, urban load-dense areas, and renewable energy connection points. In particular, data centers require large amounts of reliable power, while space for new power facilities is often limited. HTS power technology may provide a compact and high-capacity power supply option for such cases.

Industrial Players and Activities Highlights

The commercialization of HTS power technology requires cooperation among electric utilities, cable manufacturers, electrical equipment manufacturers, cooling system companies, construction companies, and operation solution providers. An HTS power system is not only a cable product. It is a combined infrastructure that includes cooling, protection and control, monitoring, construction, and maintenance.

In July 2025, KEPCO, LS Cable & System, and LS ELECTRIC signed an MOU to promote superconducting transmission business. This MOU is meaningful because it provides a cooperation framework to move HTS power technology from demonstration to practical business application.

Since then, the three companies have been working together to explore the application of HTS technology to data center power supply. Data centers require large amounts of reliable electricity, but they often face limitations in available space and grid connection conditions. HTS cables can transmit large amounts of power in a compact space, so they can be considered as a new option for data center power supply.

The three companies are also reviewing the use of HTS technology for renewable energy connection. Large-scale renewable energy projects require careful review of transmission capacity, available land, cable routes, power loss, and grid stability. HTS technology may help increase transmission capacity and reduce space requirements for such connection facilities.

In addition, the three companies are cooperating in planning future technologies such as MVDC HTS cables and SFCLs. MVDC HTS cables can be considered for renewable energy integration, industrial complexes, and urban load-dense areas. SFCLs can help reduce fault current and lower the burden on existing substations and circuit breakers.

For wider application of HTS technology to private customers, performance assurance and maintenance support will also be important. KEPCO is reviewing the use of its asset management technologies to diagnose the condition of cooling systems, circulation pumps, chillers, protection and control equipment, and connected power facilities. This can help predict maintenance and replacement timing and improve the reliability of HTS power systems in future commercial applications.

Success Stories and Lessons Learned

The Munsan-Seonyu superconducting platform project is an important step for Korea's practical use of HTS power technology. The completion of buildings, cable installation, cooling system installation, and about 85% system progress means that the main infrastructure for grid application has been largely secured.

One key lesson is that HTS power technology should be treated as a total system. Cable performance is important, but reliable operation also requires a cooling system, protection and control, monitoring, maintenance, construction quality, and a grid operation method.

Another lesson is that wider use in the private sector requires a reliable operation and maintenance framework. The performance of HTS equipment must be monitored continuously, and the maintenance timing of major components should be predicted in advance. Since it may be difficult to keep major spare parts available at all times, condition monitoring and asset management-based preventive maintenance will be important.

The project also shows the importance of field demonstration and standardization. Laboratory tests alone are not enough to verify actual grid application. Experience from installation, cooling, testing, operation, and maintenance is needed. Based on the Munsan-Seonyu project, Korea needs to prepare design models, testing procedures, operation standards, maintenance guidelines, and economic evaluation methods.

Next Term

In the next term, Korea will focus on the completion and stable commissioning of the Munsan-Seonyu superconducting platform demonstration project. The main activities will include cooling operation, electrical testing, protection coordination review, and grid connection tests toward the completion target of October 2026.

Korea will also promote follow-up project development based on the results of the Munsan-Seonyu project. Candidate areas include data centers, industrial complexes, urban load-dense areas, and renewable energy connection points.

In parallel, Korea will continue to plan future R&D topics such as 154 kV-class SFCLs, MVDC HTS cables, and asset management-based operation technologies for HTS power systems. Korea will continue to develop HTS power technology as a practical solution for future power grid challenges through demonstration, standardization, business development, industrial cooperation, and advanced operation technologies.